

Rainbow in a Glass: Density Tower

Elementary Science Lesson Plan — Density & Liquid Layers

Subject	Physical Science
Topic	Density of Liquids
Grade Level	Grades 2–5 (adaptable)
Duration	45–60 minutes
Setting	Classroom or supervised home activity

Learning Objectives

- Define density as how heavy something is for its size.
- Explain why some liquids float and others sink based on density.
- Predict and observe the order of liquid layers in a density tower.
- Connect the rainbow colors to the layered liquids.
- Practice safe lab procedures and clean-up.

Background Concepts

What is a rainbow? Rainbows are made of seven colors that appear when sunlight passes through raindrops: red, orange, yellow, green, blue, indigo, and violet.

What is density? Density describes how heavy something is for its size. A rock sinks in water because it is dense; a leaf floats because it is less dense. In this experiment, students will stack liquids of different densities to create a colorful "rainbow" tower.

Fun fact: Honey is about 1.4 times heavier than water for the same volume, which is why it always sinks to the bottom of the glass.

Materials Needed

- Honey
- Dish soap
- Water + blue food coloring
- Vegetable oil
- Rubbing alcohol + red food coloring
- Clear tall glass or jar
- Spoon or dropper
- Measuring cups

Procedure

Step 1 — Pour Honey

Carefully pour honey into the glass. Honey is the heaviest liquid, so it stays at the very bottom of the rainbow.

Step 2 — Add Dish Soap

Slowly pour dish soap on top of the honey. Pour gently so the layers don't mix. Soap is lighter than honey but heavier than water.

Step 3 — Add Colored Water

Mix water with blue food coloring and pour it on top of the dish soap, using a spoon to pour slowly so layers stay separate and neat.

Step 4 — Add Vegetable Oil

Carefully pour vegetable oil on top of the water. Oil is lighter than water, so it floats on top. Pour very slowly.

Step 5 — Add Rubbing Alcohol

With an adult's help, add rubbing alcohol colored with red food coloring on top of the oil. This is the lightest liquid and stays on top.

Result: Students should see a colorful rainbow of stacked layers in the glass — from bottom to top: honey, dish soap, blue water, oil, and red alcohol.

Why Do the Layers Stay Separate?

Each liquid has a different density, so they don't mix. The heaviest liquid sinks to the bottom, while the lightest floats on top — like stacking balls from heaviest to lightest.

Safety Guidelines

- Do NOT drink any liquids used in the experiment.
- Ask an adult for help, especially with the rubbing alcohol step.
- Clean up spills right away.
- Wash hands after the experiment.

Clean-Up

Rinse the glass and tools with warm water, put away all materials, and wipe the table clean.

Review & Discussion Questions

- What colors did you see in your rainbow tower?
- Which liquid was the heaviest? Which was the lightest?
- Why don't the liquids mix together?
- What safety rules should you remember during this experiment?
- Can you think of other examples where dense and less-dense substances separate (e.g., oil and water, ice and water)?

Extension Activity

Have students predict the order of layers before starting, then compare their predictions to the actual results. For older students, introduce the concept of density as mass divided by volume (g/mL) and have them research the approximate densities of each liquid used.

Great job, scientists! Keep exploring the world around you.